

FINDING OUT ABOUT MIGRATION



To learn how far birds migrate and where they go, scientists put metal or plastic rings on their legs. The rings are numbered or color-coded and have an address on them, like the one this juvenile white-tailed eagle is wearing. It is then possible to identify individual birds and keep records of their migration movements. Another way of tracking birds' movements over shorter distances is by using a radio collar.

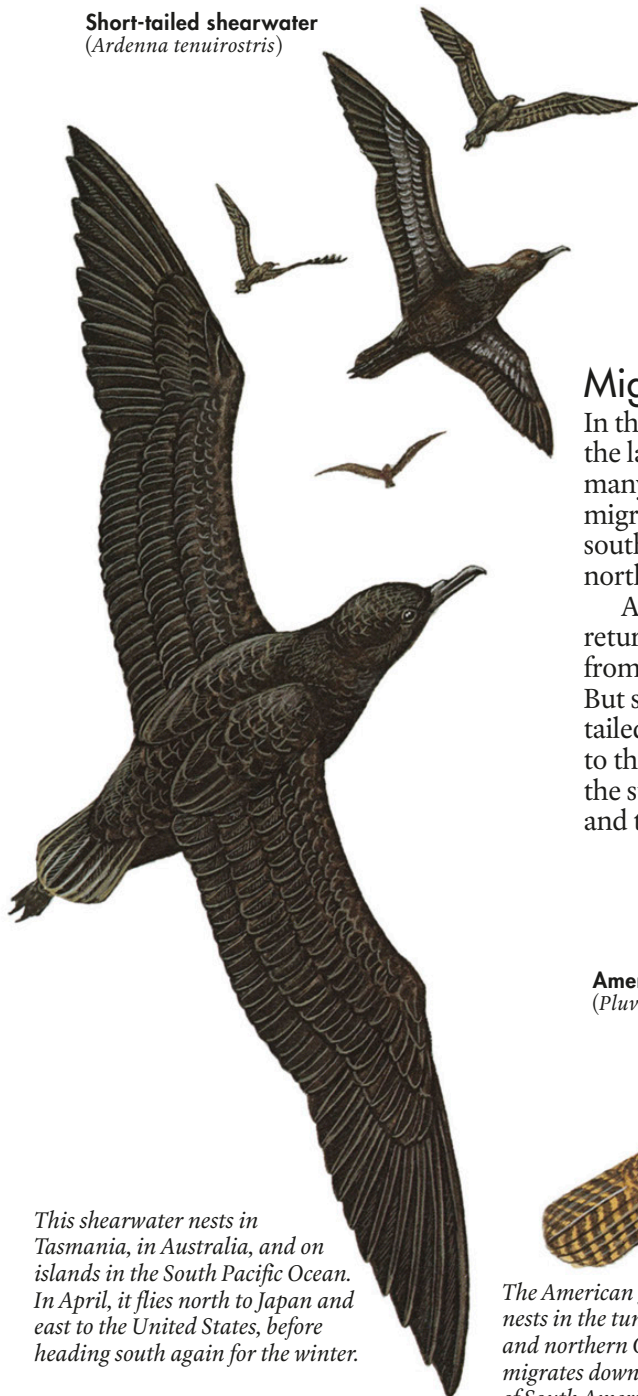
Common cranes migrate by day and by night from Europe to Africa.



Common crane
(*Grus grus*)

Birds often migrate in flocks of thousands of birds. Flying in a V-formation helps these common cranes save energy on a long journey.

Short-tailed shearwater
(*Ardenna tenuirostris*)



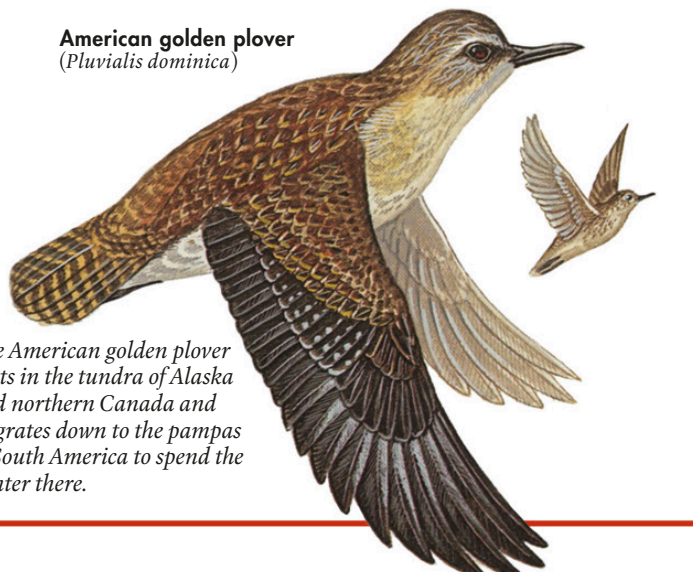
This shearwater nests in Tasmania, in Australia, and on islands in the South Pacific Ocean. In April, it flies north to Japan and east to the United States, before heading south again for the winter.

Migration from north to south

In the northern hemisphere, where much of the land is covered with ice and snow in winter, many birds, including the American golden plover, migrate. They spend summer in the north and fly south to warmer places for the winter, returning north again in spring.

Apart from these birds making their return journey, far fewer land birds migrate from the southern to the northern hemisphere. But several southern seabirds, such as the short-tailed shearwater, do migrate over the Equator to the oceans of the northern hemisphere for the summer because the climate is warmer and there is more food there.

American golden plover
(*Pluvialis dominica*)



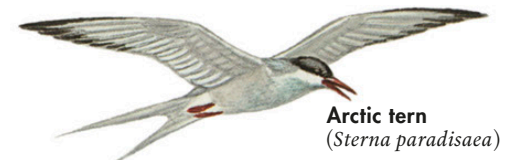
The American golden plover nests in the tundra of Alaska and northern Canada and migrates down to the pampas of South America to spend the winter there.

The journey

Birds use a lot of energy for continuous flapping flight, so on long journeys they have to try and conserve their strength. Some, such as these common cranes, fly in a V-formation so that the birds following the leader do not use up as much energy pushing aside the air as they fly. Others, such as white storks, make use of rising hot air currents—called thermals—to glide and soar without using up any energy. Seabirds glide on currents rising off the ocean waves.

Some birds can build up enough stores of body fat to fly nonstop for four days; others have to stop and feed every day.

AMAZING MIGRANTS



Arctic tern
(*Sterna paradisaea*)

Champion migrant

Every year, the Arctic tern flies from the top to the bottom of the world and back again—a distance of about 22,000 miles (36,000 km).

High fliers

Most birds fly below 300 ft (91 m) on migration, but some cross high mountains. Bar-headed geese in the Himalayas migrate at heights of more than 31,168 ft (9,500 m).

Weight-watcher

Small birds, such as the blackpoll warbler, may double their weight before migrating in order to “fuel” their journey.

Tiny migrant

Most hummingbirds do not migrate, but the ruby-throated hummingbird travels up to 2,000 miles (3,200 km) across the eastern US and the Gulf of Mexico to Central America. These tiny birds must refuel to sustain such a long flight.



Ruby-throated hummingbird
(*Archilochus colubris*)